

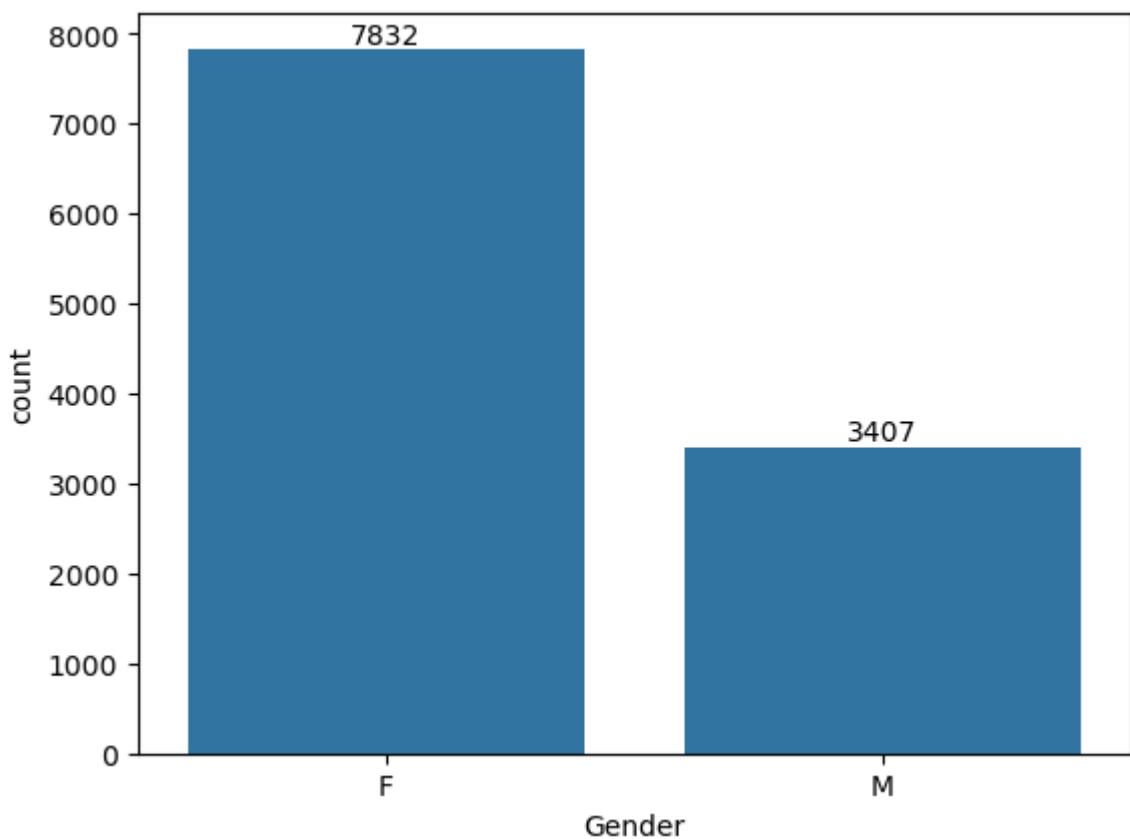
OBJECTIVE:

1. Improve customer experience by analyzing sales data
2. Increase revenue

Exploratory Data Analysis:

Gender

In [53]:

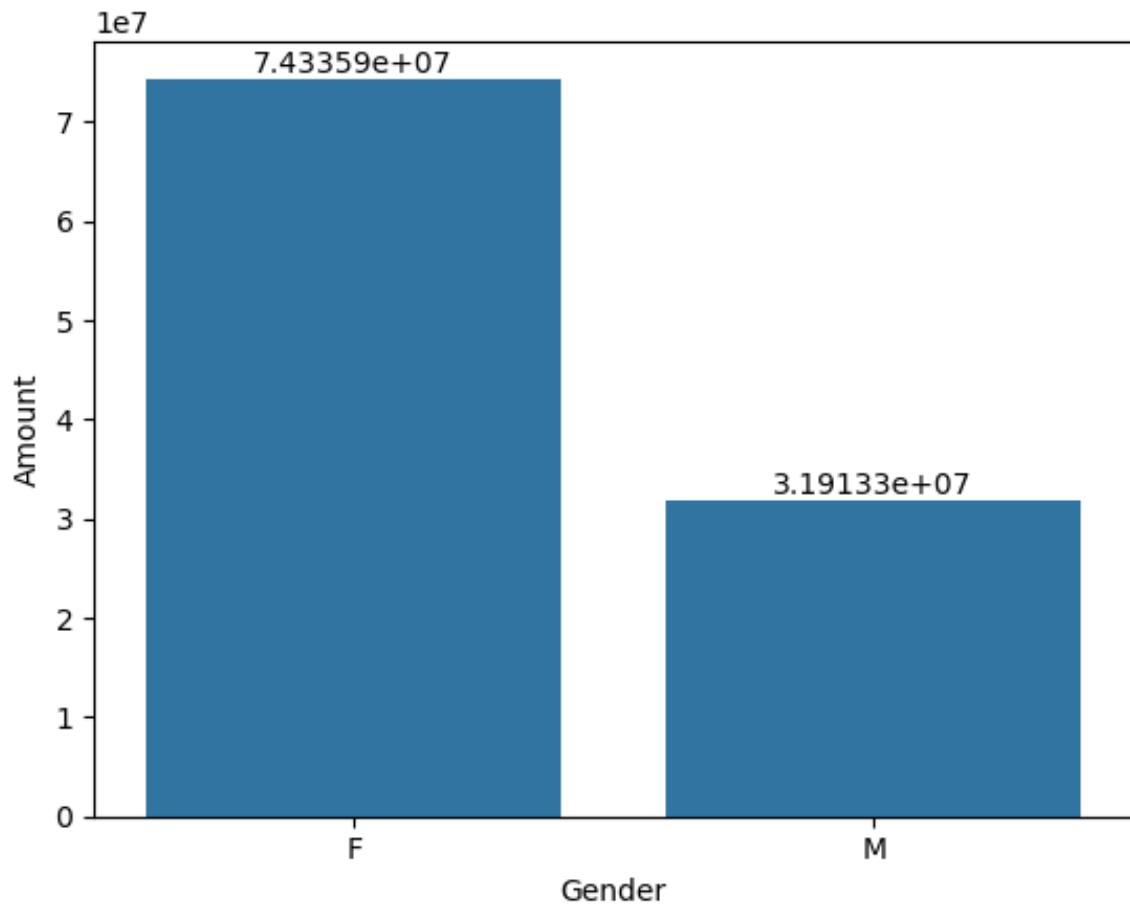


In [55]: sales_gen

Out[55]:

	Gender	Amount
0	F	74335853
1	M	31913276

In [56]:

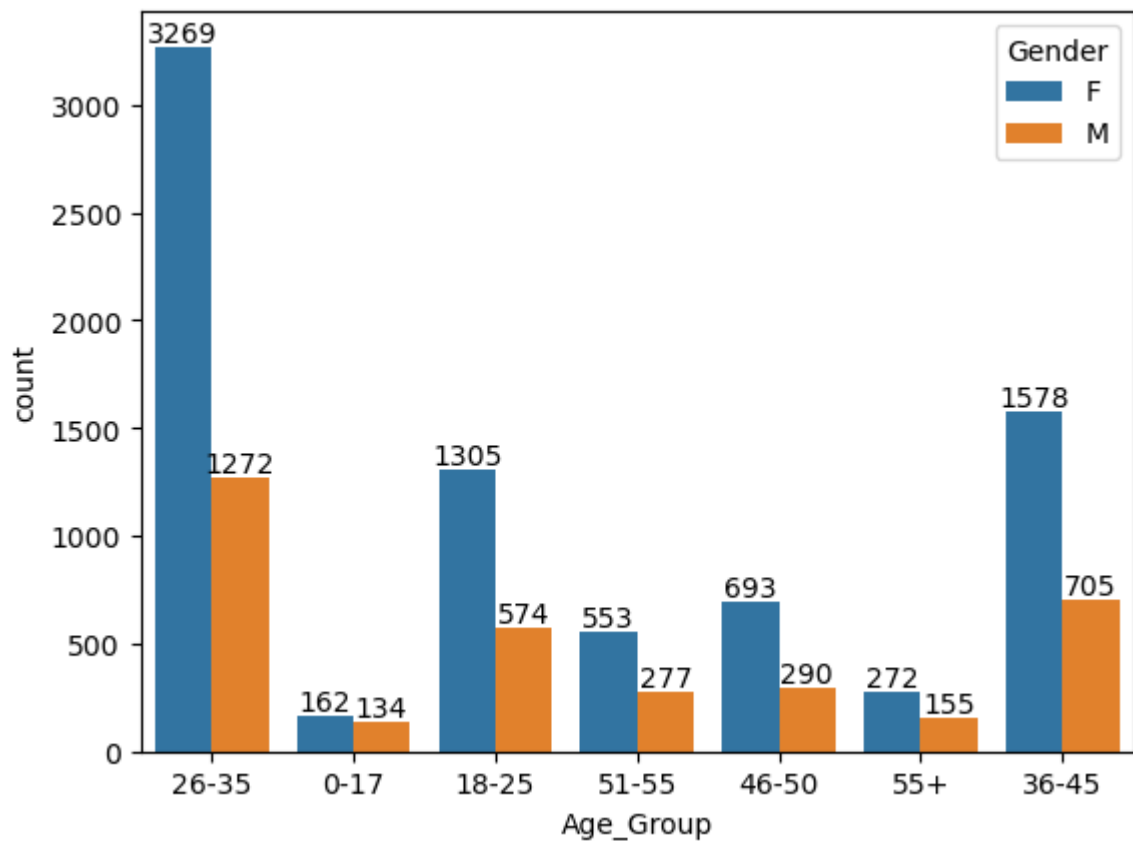


Findings:

From the above graphs we can see that most of the buyers are females and even the purchasing power of the females are greater than men

Age

In [99]:

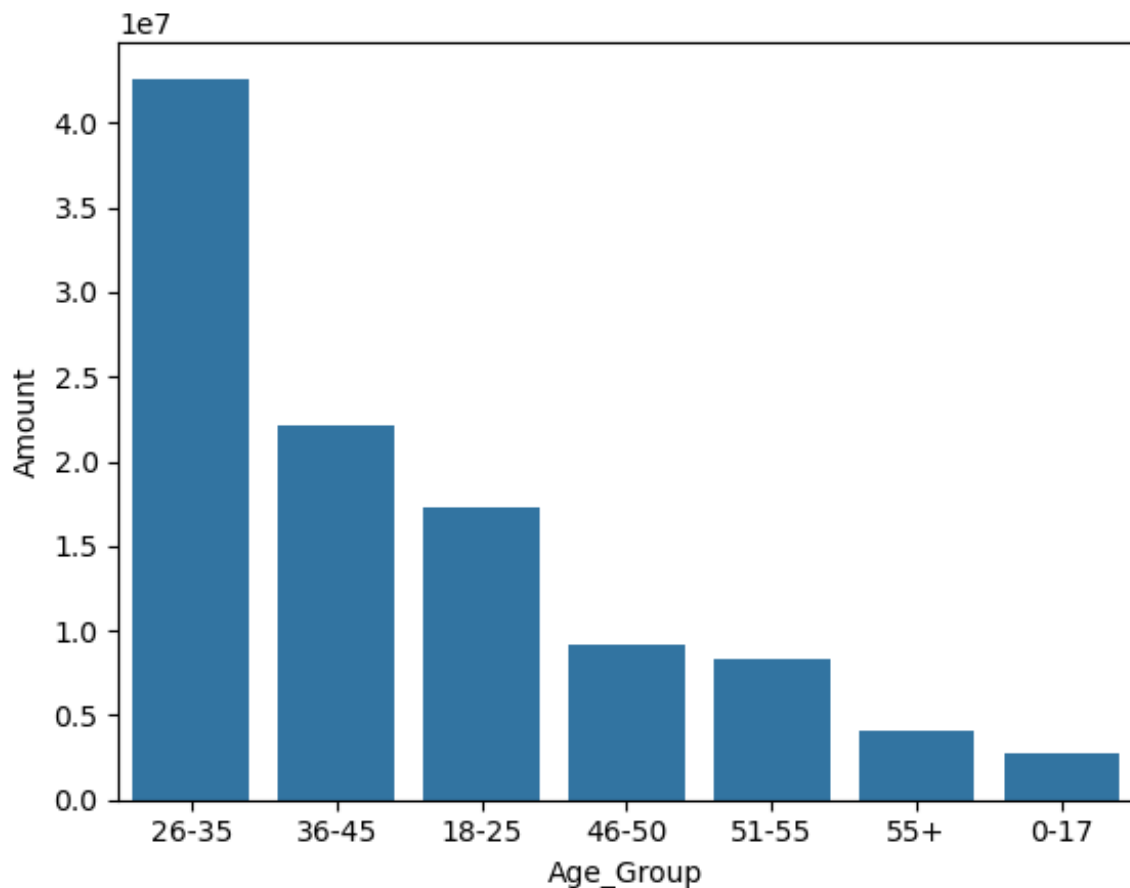


In [101...

Out[101...

	Age_Group	Amount
2	26-35	42613442
3	36-45	22144994
1	18-25	17240732
4	46-50	9207844
5	51-55	8261477
6	55+	4080987
0	0-17	2699653

In [102...



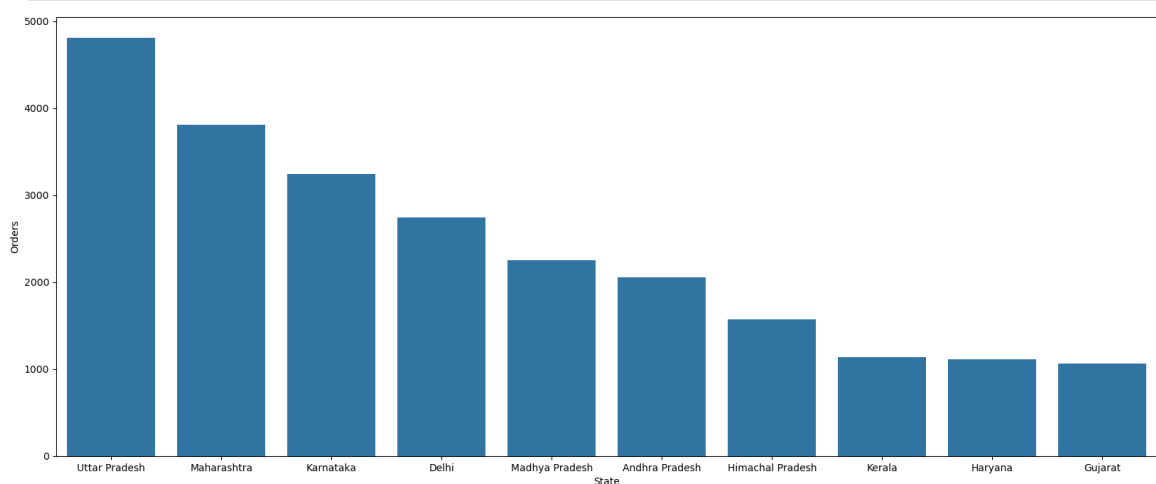
Findings:

From above graphs we can see that most of the buyers are female of age group between 26-35 years

State

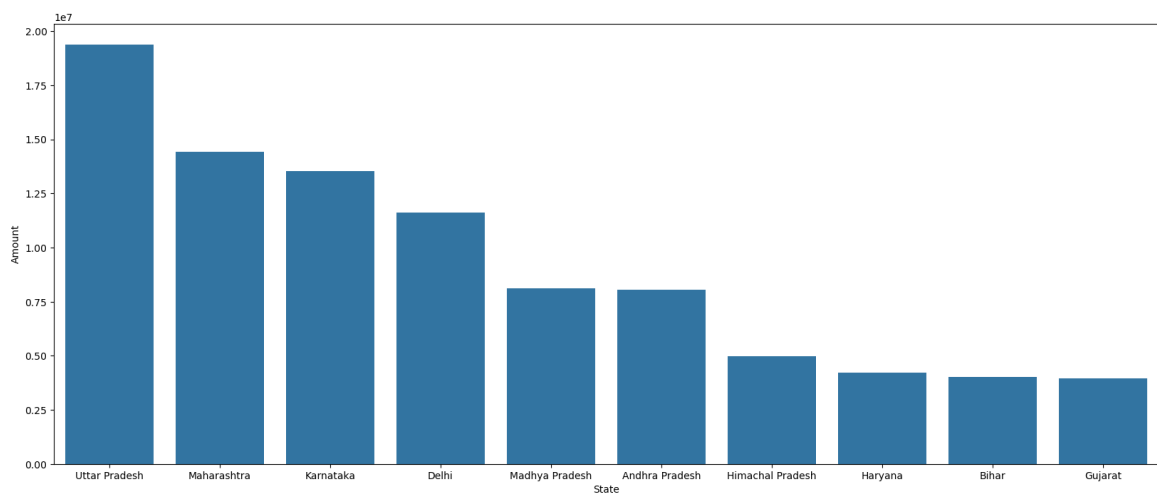
Total number of order from top 10 states

In [105...



Total amount from top 10 states

In [107...]



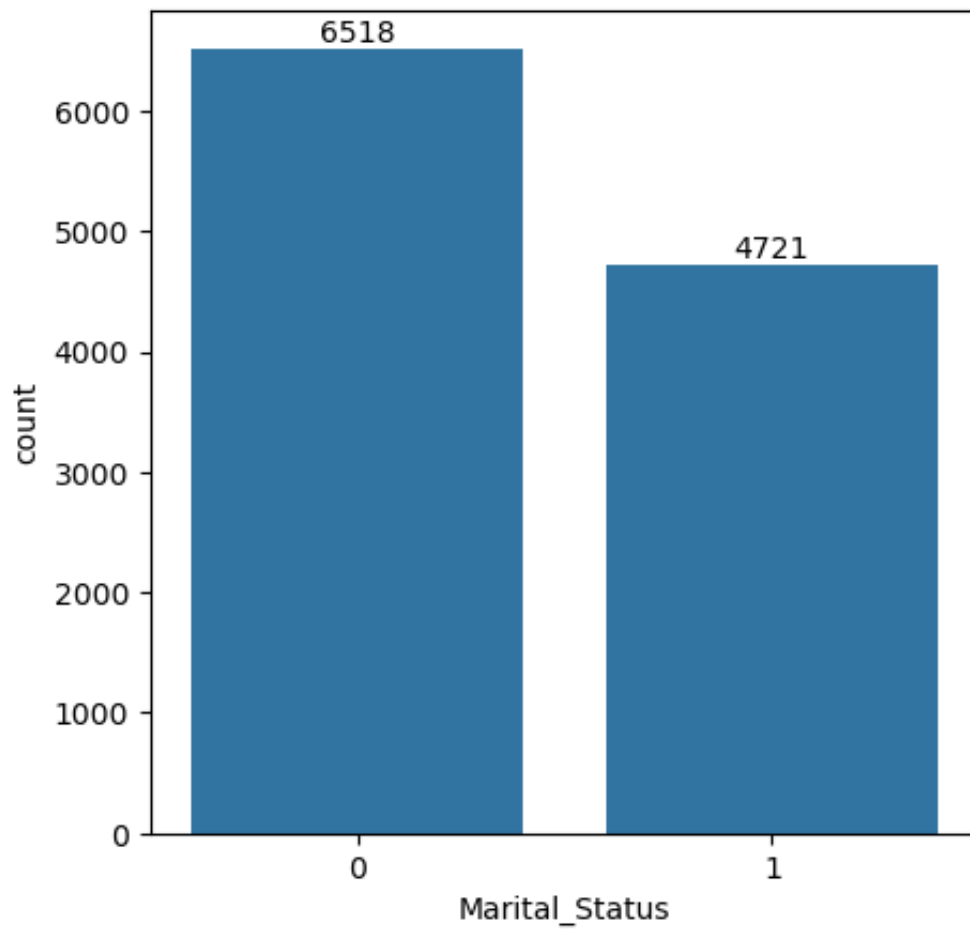
Findings:

From the above graphs we can see that most of the orders are from Uttar Pradesh, Maharashtra and Karnataka.

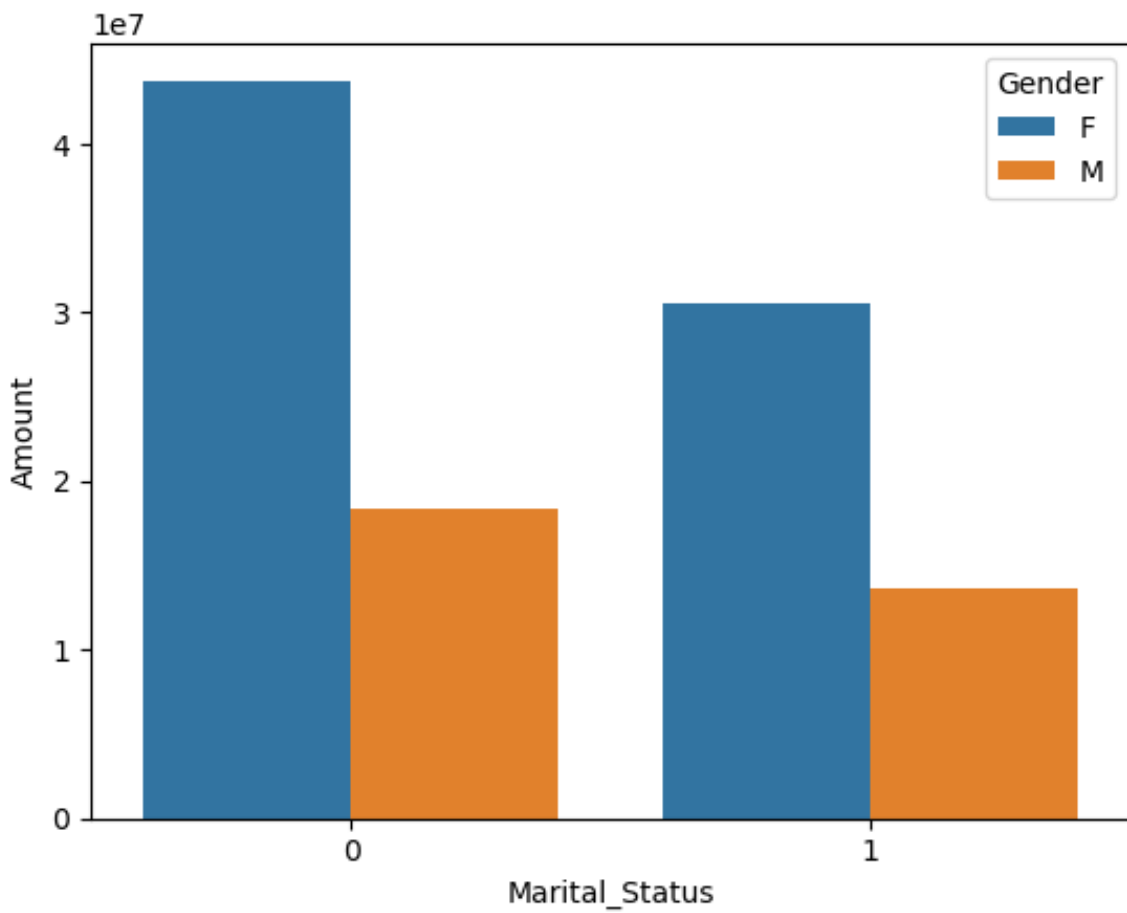
Spending power of Haryana is greater than Kerala.

Marital Status

In [109...]



In [110...

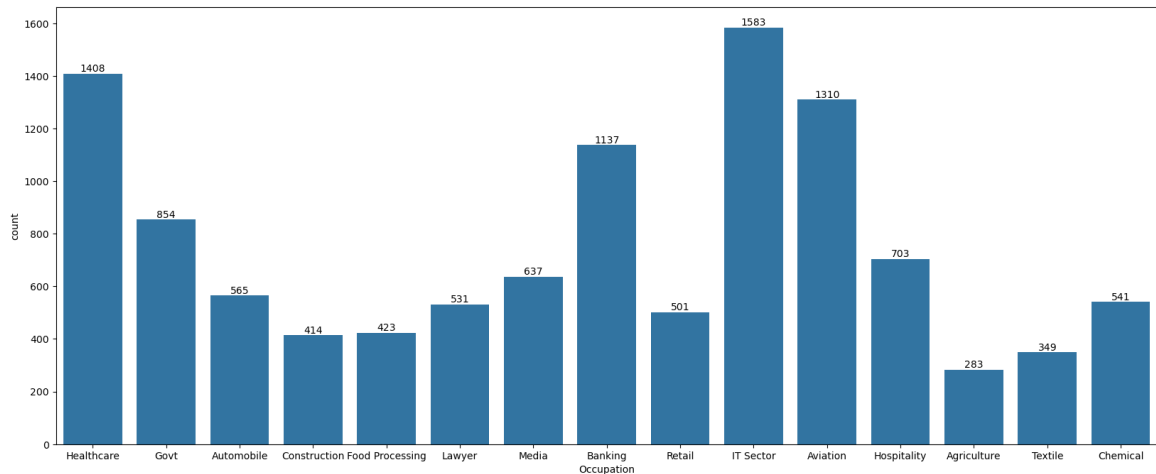


Findings:

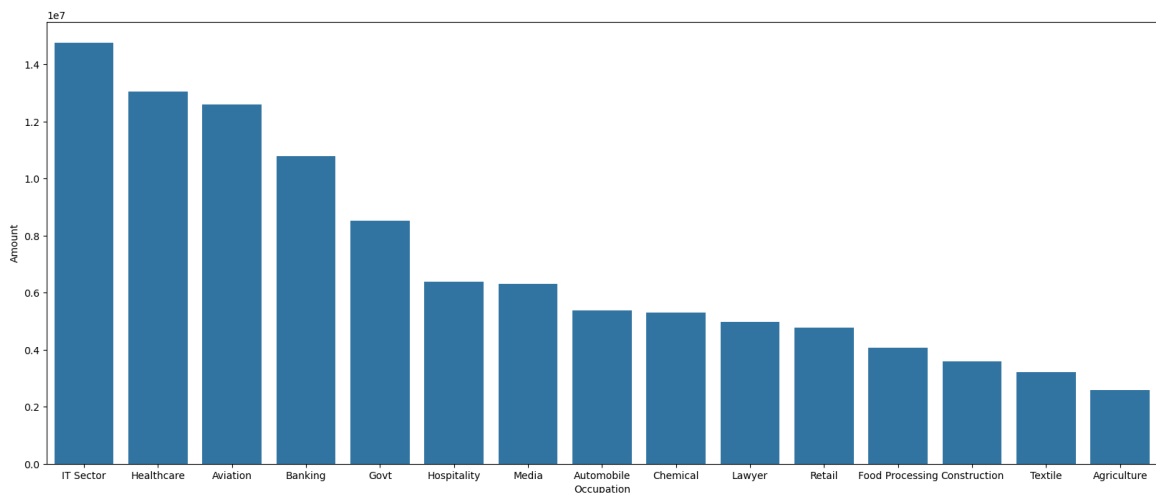
from the above graphs we can see that most of the buyers are married women and they have higher purchasing power

Occupation

In [112...



In [113...

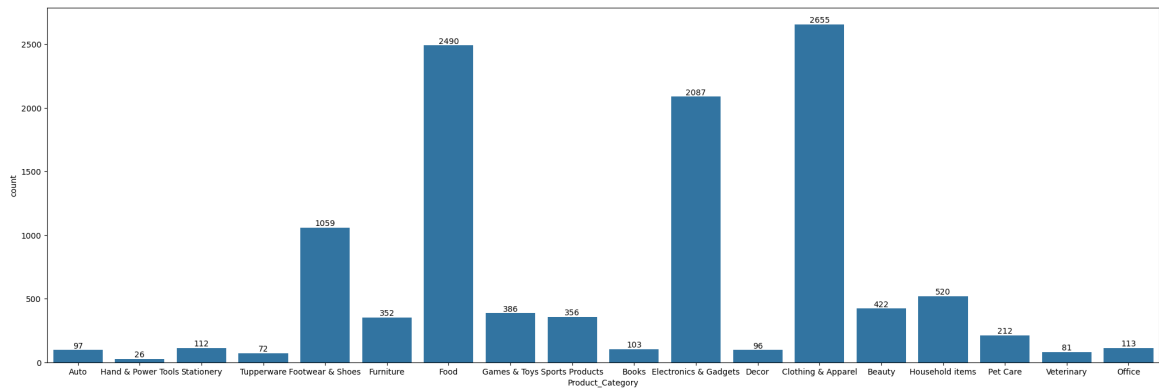


Findings:

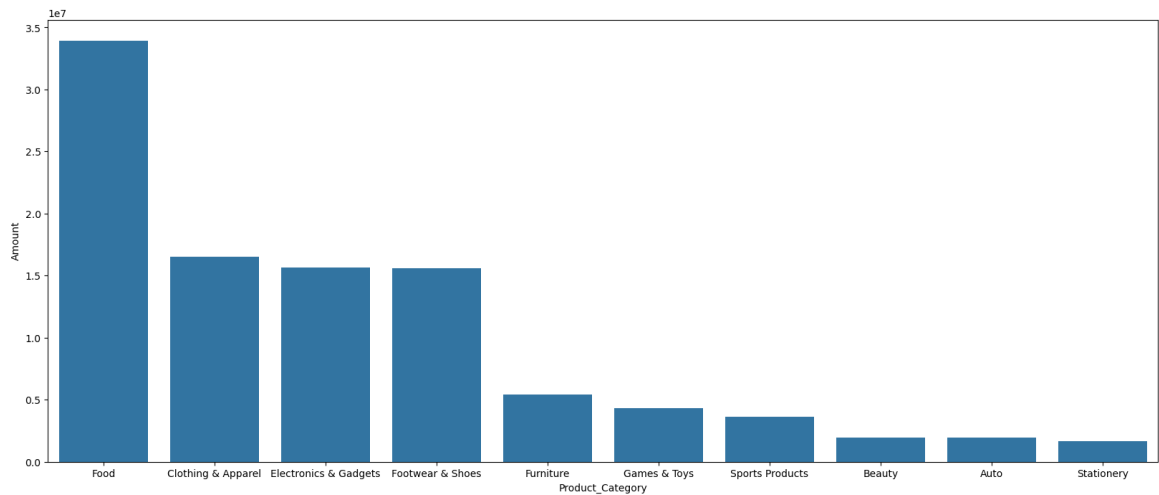
The purchasing power/ most of the buyers are working in IT, Healthcare and Aviation sector.

Product Category

In [135...



In [136...]



Findings:

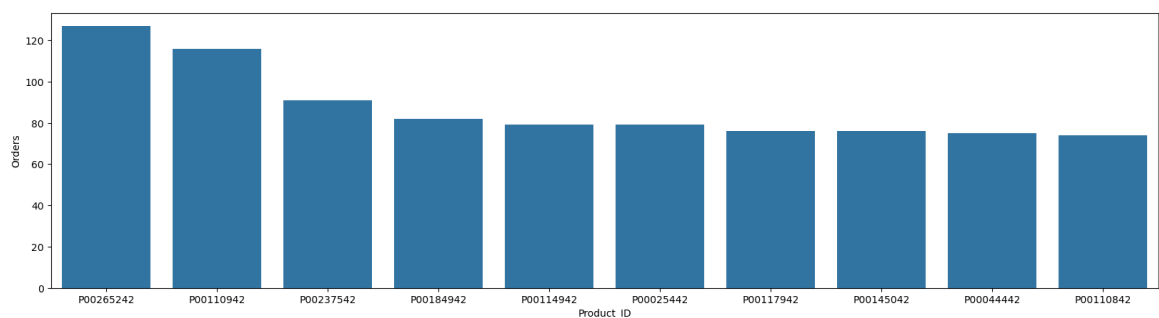
Most of the products sold are Food, Clothings, Footware and Electronics.

Purchasing power Food > Clothing > Electronics

Product ID

Top selling products

In [142...]



Customer Segmentation & Sales Analysis Report

Objective

- Improve customer experience by identifying high-value customer segments
 - Increase revenue through targeted, data-driven strategies
-

Key Findings

1. Gender Analysis

Female customers account for approximately **70% of total revenue** (₹74.3M vs ₹31.9M), clearly dominating both purchase frequency and spending.

Interpretation: Female customers are the primary revenue drivers, not just in volume but in overall value contribution.

2. Age Group Analysis

The **26–35 age group** contributes the highest revenue (~₹42.6M), followed by the **36–45** and **18–25** segments.

Interpretation: Young to mid-age working professionals form the core customer base.

3. Geographic (State-Level) Analysis

The highest order volumes are observed in:

- Uttar Pradesh
- Maharashtra
- Karnataka

However, spending patterns vary across states.

Interpretation: High order volume does not always translate to high revenue. Both demand and average spend must be considered for accurate targeting.

4. Marital Status Analysis

Married women contribute a significantly higher share of total purchases.

Interpretation: This segment represents a stable, high-value customer group with consistent purchasing behavior.

5. Occupation Analysis

Top contributing professions include:

- IT
- Healthcare
- Aviation

Interpretation: Customers from stable, higher-income professions contribute more to overall revenue.

6. Product Category Analysis

Top-performing categories by revenue:

- Food
- Clothing
- Electronics

Interpretation: Sales are driven primarily by essential and lifestyle products, indicating recurring and necessity-based purchasing behavior.

Business Recommendations

1. Target High-Value Customer Segment

Focus on:

- Females aged 26–35
- Working in IT, Healthcare, or Aviation
- Located in Uttar Pradesh, Maharashtra, and Karnataka

Action: Deploy targeted digital marketing campaigns and personalized promotions.

2. Increase Average Order Value (AOV)

Action:

- Introduce product bundling (e.g., clothing with accessories)

- Implement cross-selling strategies for electronics (add-ons, warranties)
-

3. Optimize Inventory by Region

Action:

- Increase stock levels in high-demand states
 - Improve logistics and delivery speed to enhance repeat purchases
-

4. Implement Segment-Based Pricing

Action:

- Identify regions with higher average spending
 - Introduce premium offerings in those markets
-

5. Strengthen Customer Retention

Action:

- Launch loyalty programs
 - Use personalized email and SMS campaigns for repeat engagement
-

Conclusion

The primary revenue-driving segment consists of **married women aged 26–35**, particularly those employed in stable professional sectors and located in high-demand states.

Aligning marketing, inventory, and pricing strategies with this segment will directly improve customer experience and drive sustained revenue growth.